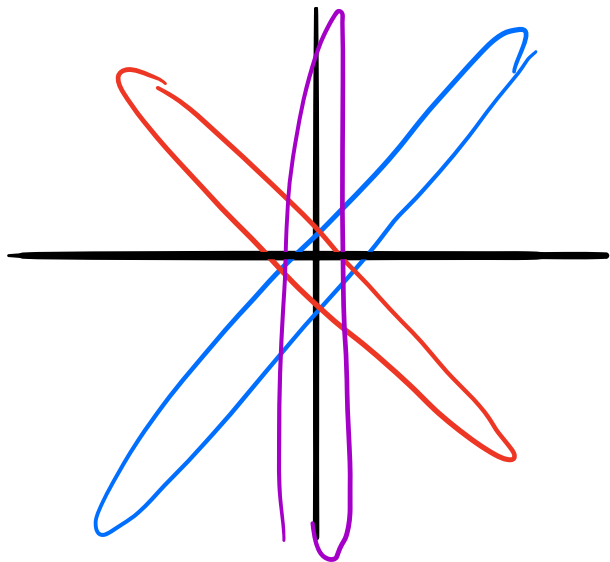


On a Deep Connection Between Simultaneous Approximations and Separation Bounds

Neta Singer

joint work with Matthieu Haeblerle, Friedrich Eisenbrand.



EPFL

$$\sum_{i=1}^n x_i \sqrt{\beta_i}$$

Two Questions in the Geometry of Numbers

Simultaneous Approximation

For $d \in \mathbb{R}$, $\delta > 0$,
is there a quantitative bound
on q satisfying:

$$\left| d - \frac{p}{q} \right| < \frac{1}{q^{2+\delta}} ?$$

Separation Bounds

How many bits of precision
are needed to decide if

$$\sum_{i=1}^n y_i \sqrt{z_i} > 0 ?$$

for $y_i \in \mathbb{Z}$, $z_i \in \mathbb{N}$

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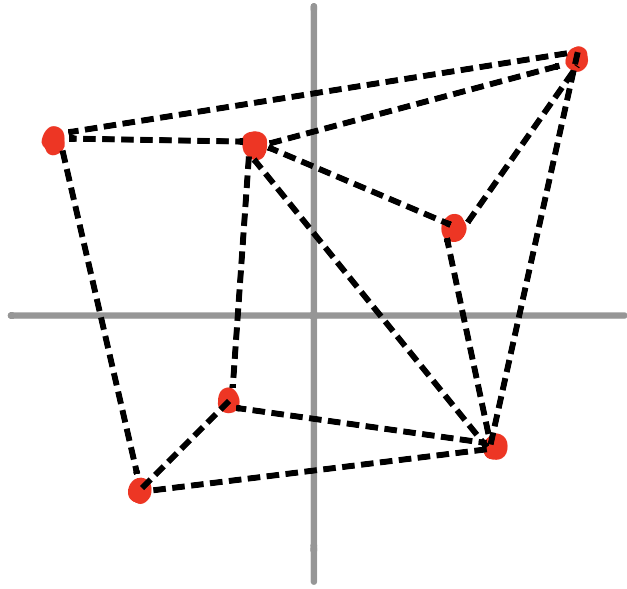
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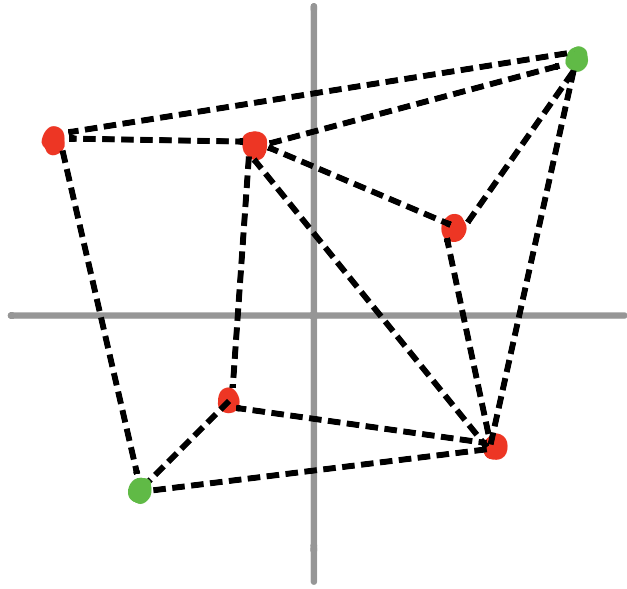
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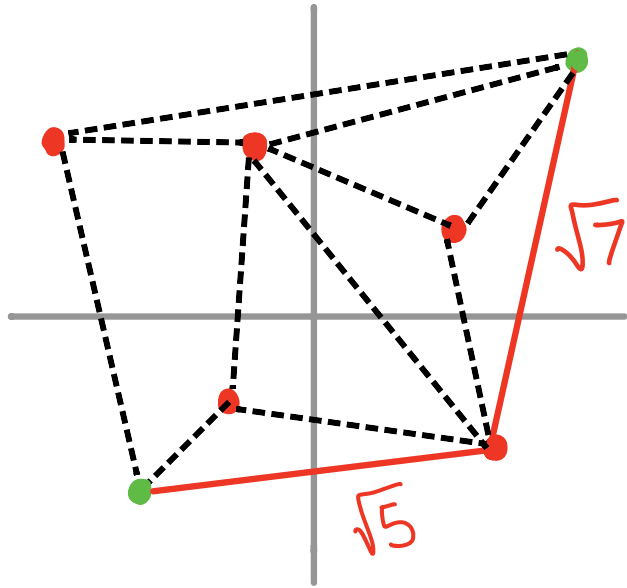
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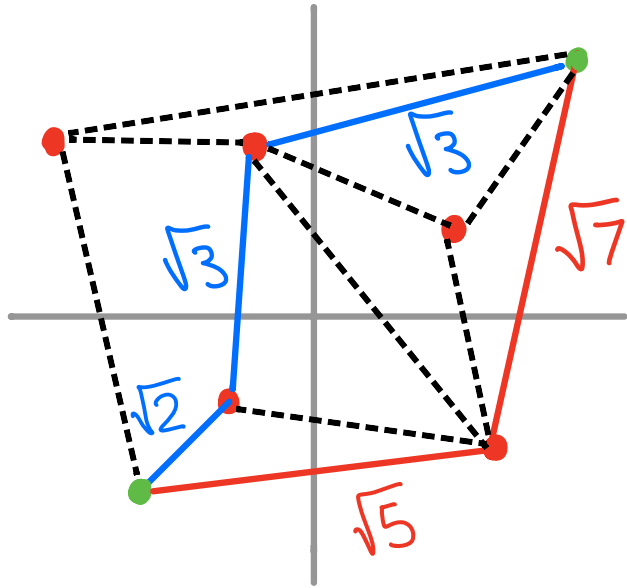
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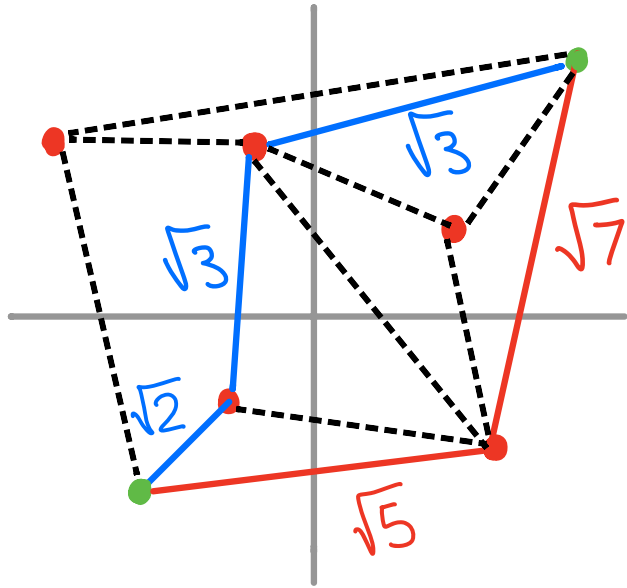
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$$|s \quad (\sqrt{7} + \sqrt{5}) < (\sqrt{2} + 2\sqrt{3})?$$

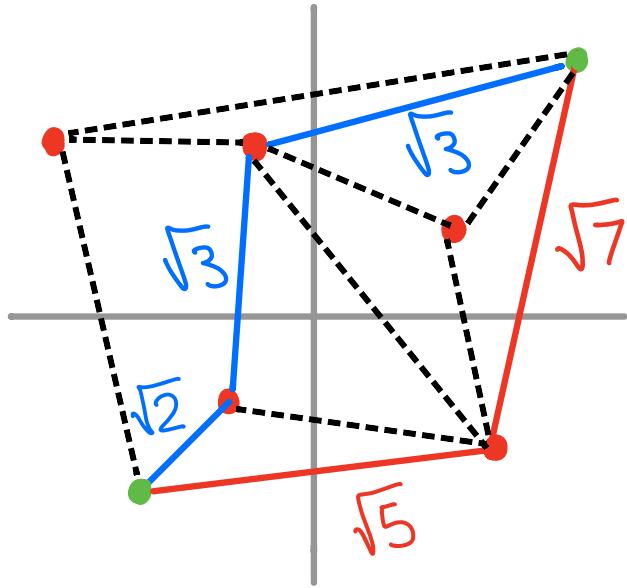
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$$|s \sum_{i=1}^k y_i \sqrt{z_i} < \sum_{i=k+1}^n y_i \sqrt{z_i} ?$$

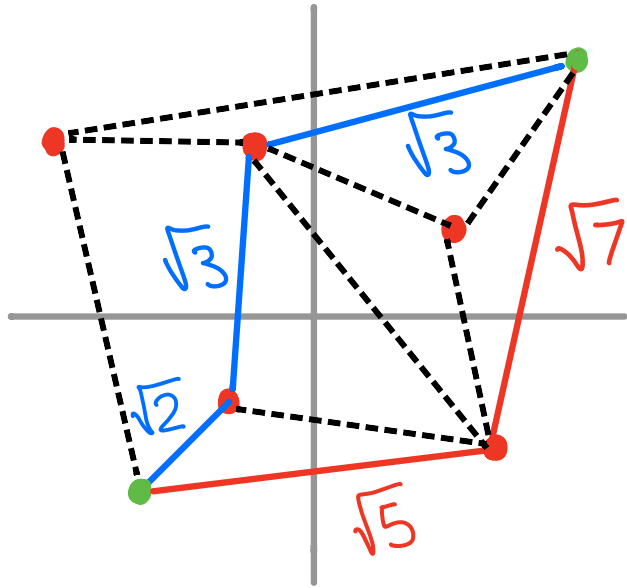
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$$\text{Is } \sum_{i=1}^n y_i \sqrt{z_i} > 0 ?$$

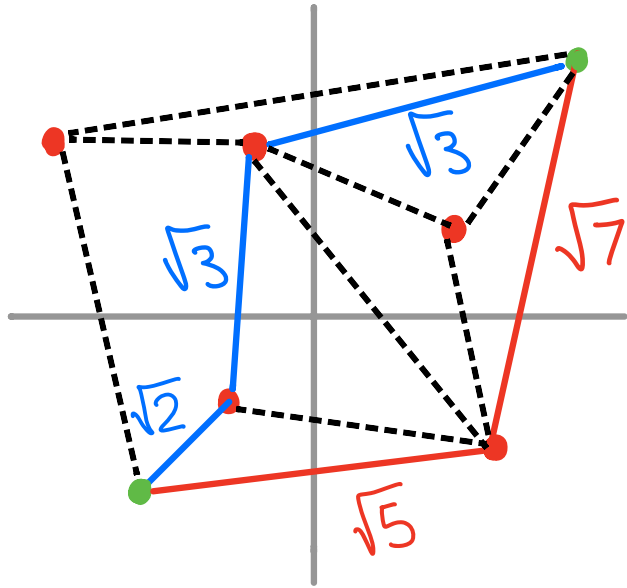
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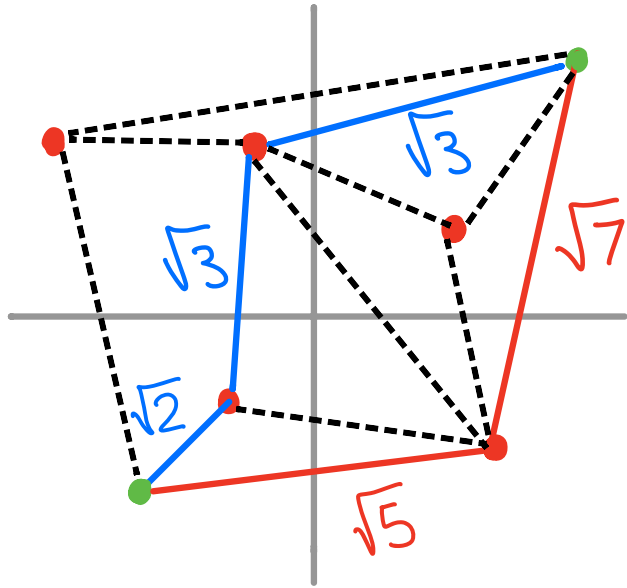
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Separation Bounds of the form $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > f(\|z\|, \|y\|, n)$

Two Questions in the Geometry of Numbers



Separation Bounds of the
form $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > 2^{-\text{poly}(n, \|z\|, \|y\|)}$

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For $d \in \mathbb{R}$, $S > 0$,
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infinitely many $p, q \in \mathbb{Z}$ such that

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Roth's Theorem (1955):

For $d \in \mathbb{R}$, $\delta > 0$, finitely many
 $p, q \in \mathbb{Z}$ s.t.

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Schmidt's Subspace Theorem (1972):

Higher dimensional variant of Roth's.

Separation Bounds for Sums of Square Roots

Prior Bound [BFMS'2000]:

$$\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > \left(n \cdot \max_i (|y_i| \sqrt{z_i}) \right)^{-(2^n - 1)}$$

Quantitative Analysis [CMSC'2010]

- Numerical approximations of $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right|$
- Approximations made via the lattice generated by

$$\begin{bmatrix} \mathbb{Q} & \lceil \mathbb{Q} \sqrt{z_1} \rceil & \lceil \mathbb{Q} \sqrt{z_2} \rceil & \cdots & \lceil \mathbb{Q} \sqrt{z_n} \rceil \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

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Quantitative Analysis [CMSC'2010]

- Numerical approximations of $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right|$
- Approximations made via the generated lattice
- Shortest lattice vector observed for given Q
- Conjecture: $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > \left(2 \cdot n \cdot \max_i (|y_i| \sqrt{z_i}) \right)^{-\text{poly}(n, \|z\|_\infty)}$

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Our Observation:

$$\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > \left(n \cdot \max_i (|y_i|) \right)^{-(2 \cdot n)} \cdot \gamma$$

Separation Bounds for Sums of Square Roots

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Techniques Used:

- Using the dual lattice Λ^* generated by $\begin{bmatrix} 1/Q^{n+1} & 0 & \dots & 0 \\ \sqrt{z_1} & 1 & & 0 \\ \vdots & \vdots & 1 & \\ \sqrt{z_n} & 0 & 0 & \ddots & 1 \end{bmatrix}$
- Subspace theorem: finitely many q satisfying

$$|q\sqrt{z_i} - \lfloor q\sqrt{z_i} \rfloor| < q^{-(1+\epsilon)\frac{1}{n}} \quad \forall i$$

- $\exists Q_0$ such that for $Q \geq Q_0$, $\lambda_1(\Lambda^*) \geq \frac{1}{Q^{1+\epsilon}}$

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Our Observation:

$$\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > \left(n \cdot \max_i (|y_i|) \right)^{-(2 \cdot n)} \cdot \gamma$$

- γ is a constant that depends only on the bound on integers q satisfying

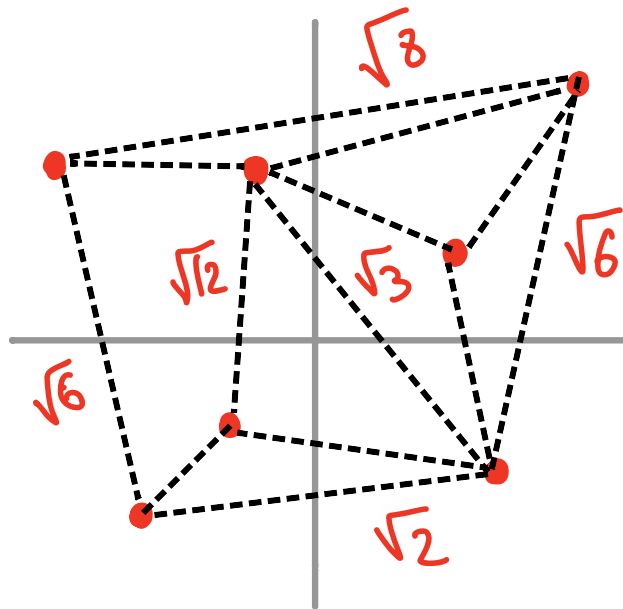
$$|q\sqrt{z_i} - \lfloor q\sqrt{z_i} \rfloor| < q^{-\frac{1}{n}(1+\epsilon)}$$

Separation Bounds for Sums of Square Roots

Our Observation:

$$\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| \geq (n \cdot \|y\|_\infty)^{-2 \cdot n} \cdot \gamma$$

- if $z_i = \prod_{j=1}^n a_j^{m_{ij}}$ for each i , then γ depends on $a_1, \dots, a_n$



$$\rightarrow \gamma = f(\sqrt{2}, \sqrt{3})$$

Separation Bounds for Sums of Square Roots

Our Observation:

Separation bounds for
sums of radicals



Upper bounds on
simultaneous
approximators

- New separation bounds $\gamma = f(q) \Leftarrow$ quantitative bound on q
- Separation bounds $\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| \geq \|y\|_{\infty}^{\text{poly}(n)} \cdot C \Rightarrow$ new quantitative bounds $q = f(C)$ [joint work with Blömer].

Open Questions:

- (1) Explicit bounds on q , denominator of simultaneous approximations (even for $\sqrt{2}, \sqrt{3}!$).
- (2) Separation bounds of the form
- $$\left| \sum_{i=1}^n y_i \sqrt{z_i} \right| > (\|y\|_\infty)^{-\text{poly}(n)} \cdot C.$$
- (3) For a fixed $\sqrt{z_1}, \dots, \sqrt{z_n}$, show that the possible coefficients $\vec{y} \in \mathbb{Z}^n$ that make the expression small is sparse.